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PI Name:	Long, Marshall Ph.D.		
Project Title:	Coflow Laminar Diffusion Flame		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Special Category:	None		
PI Email:	marshall.long@yale.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-432-4229
Organization Name:	Yale University		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	PO Box 208284		
PI Web Page:			
City:	New Haven	**State:**	CT
Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
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Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
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Task Description:	Editor's Note: Successor agreement to Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A), per D. Stocker, NASA Glenn Research Center. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	Microgravity Experiments: SLICE and CLD Flame The Structure and Liftoff in Combustion Experiment (SLICE) was carried out on board the International Space Station (ISS) early in 2012. The goal of SLICE was to investigate the structure of lifted and sooty coflow diffusion flames in a microgravity environment. The SLICE experiments provided a comprehensive data set with a variety of test conditions, including different hydrocarbon fuels with varying dilution levels, different fuel nozzle sizes and various flow conditions. SLICE was a precursor to the Coflow Laminar Diffusion (CLD) Flame experiment, which is one of the experiments in the Advanced Combustion via Microgravity Experiments (ACME), slated to begin in late 2016. As will be detailed in the following sections, we continue to analyze results from SLICE in preparation for CLD Flame and ACME. Application of High Dynamic Range (HDR) Imaging Among the thousands of images of microgravity coflow diffusion flames returned by the SLICE campaign, roughly one sixth of them were partially saturated. Microgravity flames, in fact, have been found to be sootier than their 1 g counterparts and saturated images, by themselves, are not useful for quantitative measurements. The High Dynamic Range (HDR) imaging technique that has been under study to exploit the information embedded in those saturated frames, was tested and successfully applied to temperature measurements using thin filament pyrometry (TFP) and soot color ratio pyrometry in non-sooty and sooty flames, respectively. Additional characterization of the SLICE Nikon camera was done to assess reciprocity and linearity of the detector over an extended dynamic range. Representative sooty flames from the SLICE database were selected as test cases and the flame temperature was inferred using soot color ratio pyrometry, while the soot volume fraction was quantified thanks to an absolute light calibration. Results showed that the HDR approach was able to return images with enhanced signal-to-noise ratio (SNR), which in turn resulted in better-defined temperature and soot volume fraction spatial distributions. This work was presented at the 9th US Combustion Meeting and an Applied Optics paper was published. The same linearity and reciprocity analysis done on the SLICE detector was also performed on the color camera selected for the ACME experiments. It was seen that for the ACME camera, pixel cross talk and blooming become significant when images are partially saturated, and specific measures must be taken to avoid corruption of the quantitative data if the HDR approach is implemented. HDR imaging was also extended to Rayleigh measurements in time-varying/turbulent flows. Two sequential laser pulses were used as illuminating sources and the Rayleigh scattered light was imaged and recorded using the double frame option of an interline transfer CCD camera. Laser energies were such that the first image covered the majority of the detector dynamic range without reaching any saturation, while the second one was partially saturated in order to increase the pixel counts of the low signal level regions; the inter-pulse separation was chosen to be smaller than the characteristic time scales of the phenomena. The reconstructed HDR image showed increased SNR when compared to the unsaturated one, and equal or better SNR when compared to a conventionally averaged image. Current efforts are focusing on an alternative way of increasing the SNR in laser-based imaging experiments when image intensifiers are used as gating devices. The Rayleigh-scattered light of a single laser pulse is recorded and imaged on the chip of an interline transfer CCD camera through the use of an image intensifier; by exploiting the long temporal decay of the intensifier's phosphors, when compared to the laser pulse, the phosphors' emitted light can be recorded over the two consecutive frames of the CCD. Depending on the frames' exposures and their timing with respect to the phosphor decay curve, the recorded counts in the two frames can be made equal, allowing for an averaging of the two. Preliminary results showed that, when working in the linear regime of an intensifier, the SNR of the averaged image is higher that the one of a conventional Rayleigh result obtained with an equal laser energy. Quantitative Measurements of CH* Concentration As part of the ongoing analysis of the SLICE image database, we shifted our focus from temperature and soot volume fraction measurements in sooty laminar coflow diffusion flames to the study and quantification of the chemiluminescent CH* radical in the non-sooty ones. Chemiluminescence is often considered as a practical marker for flame heat release

rate and the comparison with results obtained from numerical simulations of SLICE flames let us study the relation between the chemiluminescent emission and flame heat release rate. The spectral characterization of the DSLR color camera used to acquire the flame images allowed the signal collected by the blue channel to be considered representative of the CH* emission of the A2 delta --> X2 Pi transition centered around 431 nm, while the analysis of the spectral emission of a previously characterized nitrogen-diluted laminar diffusion methane flame accounted for the contribution of chemiluminescence from emitting species other than CH*. An Abel deconvolution of the line-of-sight chemiluminescence was used to obtain the intensity distribution in a cross section of the flame and a quantification of the CH* concentration was possible thanks to an absolute light intensity calibration. Comparisons with numerical results display reasonably good agreement between measured and computed flame shapes, and showed that the integrated CH* concentration is proportional to the integrated computed heat release rate. The two-dimensional CH* and heat release rate spatial profiles match in a satisfactory way, albeit with differences in gradients and intensity distributions.

CLD Flame Burner Tests in a Sealed Chamber

Unlike the previous SLICE experiments that were performed in a vented vessel (Microgravity Science Glovebox), the upcoming ACME CLD Flame experiments will be carried out in an enclosed environment (Combustion Integrated Rack). Tests have been performed using a copy of the CLD Flame burner placed inside a sealed chamber to assess the influence of increasing pressure on the sooting behavior (thus flame luminosity) of coflow diffusion flames. Given the fuel and oxidizer mass flow rates, and over the maximum two-minute duration of ACME test points, the pressure is expected to increase by no more than 30 KPa. In normal gravity, a 25 KPa pressure difference showed that the flame luminosity, for the case of partially sooty methane flames, could change by a factor of four. This characterization, despite being limited to a few representative test cases, will aid in choosing and adjusting the optimal exposure times during the two minute image acquisition period. Additionally, thermocouple temperature measurements inside the chamber provided a sense for what the ambient temperature surrounding the flame would be if the CLD Flame burner were tested inside the ground copy of the Combustion Integrated Rack.

Recent Computational Progress

Flames from the SLICE experiments have been investigated computationally using a formulation based on the MC-Smooth vorticity-velocity formulations of the governing equations. In the physical model, the chemically reactive mixture is treated as Newtonian with diffusion assumed as Fickian. The gas-phase chemistry is described with 42 species and 250 reactions, and it is based on the GRI 3.0 mechanism, with all the nitrogen-containing species removed, except N2. The axisymmetric structure of the flames allowed the implementation of a two-dimensional computational domain with boundary conditions chosen to match the experimental ones. The computational domain extends 4.29 cm in the radial direction and 12.20 cm in the axial direction, and it is spanned by a nonuniform tensor-product grid containing 167 x 289 points. The governing equations are discretized on this grid using a nine-point computational stencil. The resulting strongly coupled, highly nonlinear equations are solved simultaneously using a damped, modified Newton's method with nested Bi-CGSTAB linear solver. Initial computations concentrated on the sooting behavior of the flames, with results published in the Proceedings of the Combustion Institute. More recently, we have turned our attention to the analysis of CH* concentration and flame heat release rate. A paper on that topic is in preparation.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
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